Phase D2 Second-Quarter Synthesis and Third-Quarter Preview

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

This note is the synthesis marker for the second quarter of Phase D2 in the CKS derivation series. D2.01 through D2.49 are complete; D2.50 marks the halfway point of the approximately 80-note Phase D2 operational-variant phase. The note performs three functions: it organizes what D2.31–0049 added beyond the first-quarter baseline in five coverage categories; it provides a coverage completeness assessment mapping each of Paper 3's D1 sub-commitments to one of three coverage states (comprehensive, near-comprehensive, or partial with third/fourth-quarter targeting); and it previews the four substantive topic areas that D2.51–D2.80 will formalize. The central finding is that at D2.50 the operational prior-art foundation for Paper 3's six claims is substantively in place. The remaining 30 notes add depth, cross-claim synthesis, and cross-paper boundary cases rather than fundamentally new prior-art territory.

1. Position and Function of This Note

D2.50 is a synthesis note, not a primary derivation note. Its function is navigational and analytical: it records what the Phase D2 operational-variant layer has established at the halfway mark and what it targets for completion.

Phase D2 derives operational variants from the approximately 30 sub-commitments that Phase D1 decomposed from Paper 3's six named claims. The pattern follows Series B: each D1 sub-commitment generates approximately three to eight operational notes at D2, covering how the sub-commitment functions under varied conditions, at varied scales, and in combination with other sub-commitments. Phase D2's approximately 80 notes span note numbers #496 through #575 in the continuous CKS derivation numbering.

The first quarter (D2.01–D2.30) established baseline operational coverage for the core sub-commitments of Claims 1 through 4: shared-substrate construction and dissolution mechanics, FAI event execution (aspect exchange, full-merge semantics, three pattern variants, exchange bounding), the three-tier conflict-handling sequence (preserve, resolve via orchestration, escalate), and the four-locus evolution-feed hand-off. That baseline is the foundation on which the second quarter builds.

The second quarter (D2.31–D2.49) does not simply add more notes in the same register. It completes coverage in five distinct directions that the first quarter left open: population-scope operations, portfolio and scale governance, quality and standards, foundational architectural confirmations, and governance rights and sovereignty. These five directions are described in §2.

2. What the Second Quarter Added: Five Coverage Categories

2.1 Population-Scope Operational Coverage

D2.31 and D2.32 formalize the two operational mechanics that underlie Claim 6's population-scale collective evolution. D2.31 covers how individual Selves configure their network participation — the governance choices each Self makes about which FAI relationships to enter, under what conditions, and subject to what standing constraints within their home perimeter. D2.32 covers pattern propagation mechanics — how governance patterns and evolved substrate content propagate through the network via governed absorption and re-contribution cycles across successive FAI events. Together these two notes establish that Claim 6's collective evolution is not emergent in the uncontrolled sense: it proceeds through governed accumulation of discrete FAI events, with each propagation step subject to the receiving Self's home-perimeter governance authority. This operational grounding of Claim 6 was absent from the first quarter, which treated population scope as a compositional consequence of Claims 1–5 rather than as a target for operational prior-art coverage in its own right.

2.2 Portfolio and Scale Governance

D2.33 through D2.35 address a gap the first quarter left: single-event FAI governance is well-covered by D2.01–D2.30, but the governance infrastructure for sustained, multi-event FAI relationships at organizational scale requires its own operational prior art. D2.33 formalizes parallel FAI event governance — how a Self manages simultaneous FAI relationships with distinct counterparties under a single home-perimeter governance configuration. D2.34 formalizes cross-organizational governance agreements — the substrate-carried terms that persist across individual FAI events and govern the ongoing relationship rather than a single instance. D2.35 formalizes governance health indicators — the observable signals that a multi-event FAI relationship is operating within its governance configuration rather than drifting toward ungoverned habituation. These three notes establish that portfolio-scale FAI governance is an extension of the single-event governance mechanics rather than a categorically different regime, and they close the prior-art gap between one-off FAI execution and ongoing inter-Self coordination relationships.

2.3 Governance Quality Standards

D2.36 through D2.39 formalize what distinguishes high-quality from low-quality FAI governance and establish the compliance floor below which a system is not operating within the Paper 3 architectural commitments. D2.36 formalizes documentation standards — the minimum substrate-carried record required to make an FAI event inspectable and reproducible. D2.37 formalizes minimum viable governance — the least-governed configuration that still satisfies Paper 3's substrate-content and human-authority commitments, distinguishing genuinely minimal governance from non-compliant shortcuts. D2.38 formalizes versioning and amendment history — how governance configurations and their changes are recorded as substrate content, maintaining the audit trail Paper 1's traceability commitment requires at the FAI scope. D2.39 formalizes post-mortem governance — the retrospective process by which FAI events become inputs to governance-configuration improvement rather than isolated executions. Together these four notes provide the quality-standards layer that converts Paper 3's architectural commitments into operationally testable claims: a governance implementation either meets these standards or falls short in a specific and identifiable way.

2.4 Foundational Architectural Confirmations

D2.40 through D2.44 confirm how Paper 1 and Paper 2 foundational commitments carry through at the inter-Self scope that Paper 3 addresses. These notes are confirmations rather than extensions: they close prior-art gaps that arise when a reader might argue that Paper 3's inter-Self framing creates exceptions to or weakens the underlying commitments.

D2.40 confirms that the instinct/reasoning separation from Paper 2 holds at FAI scope — LLM weights and instinct-layer content do not exchange through the shared substrate regardless of the governance configuration of the FAI event. D2.41 formalizes time-sensitive governance — how time constraints on FAI execution interact with the governance-configuration requirements, establishing that urgency does not create authority to bypass human-authored governance rules. D2.42 formalizes scope boundaries — the precise conditions under which a coordination interaction falls within Paper 3's inter-Self scope versus Paper 2's intra-Self scope, closing the boundary ambiguity that the two-Self minimum and the one-Self self-derived combination case would otherwise leave open. D2.43 formalizes the non-delegation principle at inter-Self scope — that governance authority over shared-substrate configuration cannot be delegated to the AI systems participating in an FAI event, even under governance rules those AI systems help administer. D2.44 formalizes domain-agnosticism — that Paper 3's architectural commitments hold regardless of the domain either participating Self serves, and that domain-specific constraints are governance-configuration inputs rather than architectural modifications.

2.5 Rights and Sovereignty Coverage

D2.45 through D2.49 formalize the governance rights that protect each participating Self's sovereignty within the inter-Self coordination architecture. D2.45 formalizes dispute resolution — the governed process by which disagreements about shared-substrate content or FAI event outcomes are handled without either Self unilaterally modifying content outside its own home perimeter. D2.46 formalizes exit rights — the conditions and mechanics by which a Self terminates an ongoing FAI relationship while preserving its home-substrate integrity and satisfying the governance-configuration terms of the relationship. D2.47 formalizes asymmetric-Self participation — how FAI events proceed when participating Selves have significantly different home-perimeter governance configurations, capability profiles, or substrate maturity levels. D2.48 formalizes single-aspect contribution — the operationally minimal FAI participation case, where one Self contributes a single aspect rather than a multi-aspect portfolio, establishing that this is valid under Paper 3's n-ary and aspect-as-unit commitments. D2.49 formalizes content-tier boundaries — the governance rules that determine which substrate content from a Self's home perimeter is eligible for FAI contribution versus retained as home-perimeter-only content, operationalizing the exchange-bounding commitment that Paper 2's instinct/reasoning separation extends at the inter-Self boundary.

3. Coverage Completeness Assessment at D2.50

The following table maps Paper 3's D1 sub-commitment clusters to their coverage state at the D2.50 midpoint. The three states are: **Comprehensive** (operational prior art established across the main variant space; no significant gap expected); **Near-comprehensive** (primary variants covered; depth or cross-claim synthesis targeted in Q3/Q4); and **Partial** (foundational coverage in place; substantive operational expansion targeted in Q3/Q4).

D1 Sub-commitment cluster	Parent claim	Coverage state at D2.50
Construction-as-temporary; perimeter-spanning property; dissolution mechanics; persistence policy as substrate content	Claim 1	Comprehensive
All six Paper 1 commitments holding within shared-substrate scope	Claim 1	Near-comprehensive (cross-claim synthesis with Claims 3 and 5 targeted)
Aspect as exchange unit; n-ary cardinality; full-merge default	Claim 2	Comprehensive
Three pattern variants; exchange bounded to substrate content	Claim 2	Comprehensive
Three persistence loci; FAI as inter-Self analog of Paper 2 intra-Self combination primitive	Claim 2	Near-comprehensive (multi-tier evolution interaction targeted)
Preserve as substrate-level state default; resolve via orchestration with inspectable rules	Claim 3	Comprehensive

D1 Sub-commitment cluster	Parent claim	Coverage state at D2.50
Escalate to humans across joint authority; cross-perimeter escalation as architectural commitment	Claim 3	Near-comprehensive (configuration-governs-conflict-handling synthesis targeted)
FAI dissolution hand-off; per-mechanism feed structure	Claim 4	Comprehensive
Three-case layer-routing rule; instinct-takes-no-FAI-input commitment	Claim 4	Comprehensive
Asymmetric ingestion sourced from per-perimeter governance	Claim 4	Near-comprehensive (multi-level directed selection interaction with Paper 2 targeted)
All configurable dimensions as substrate content; configuration-of-configuration as substrate content	Claim 5	Comprehensive
Recursive applicability; recursion bottoming at human-authored authority	Claim 5	Near-comprehensive (inter-claim synthesis: configuration governing conflict tier selection targeted)
Joint authority across population-level perimeters; accumulated FAI events as evolution mechanism	Claim 6	Near-comprehensive (additional operational aspects targeted)
Six-rung substrate-mediator ladder; calibrated-humility register; biology-resonance positioning	Claim 6	Partial (population-scope operational expansion targeted in Q3/Q4)

The coverage pattern shows a clear gradient: Claims 1 through 3 and the core of Claim 5 are comprehensively covered. The evolution-feed mechanisms of Claim 4 and the upper operational layer of Claim 6 carry the largest remaining coverage obligations. This reflects the derivation structure: Claims 1 through 3 are foundational and have relatively self-contained operational variants, while Claims 4 and 6 involve interactions with Paper 2's intra-Self machinery that require cross-paper operational notes absent from Phase D2's scope and targeted instead in Phases CC.1 and CC.2.

4. Third and Fourth Quarter Preview (D2.51–D2.80)

4.1 Inter-Claim Operational Synthesis

These notes formalize how multiple Paper 3 sub-commitments interact operationally — cases where the operational behavior of one claim is governed or constrained by another claim's mechanisms, and where the interaction produces prior art not captured by either claim's standalone operational coverage.

The primary inter-claim interactions targeted are: (a) how Claim 5 (configuration as substrate content) governs Claim 3 (three-tier conflict handling) in practice — specifically, the mechanics by which governance configuration determines at which tier a conflict is handled and how configuration versioning affects conflict-resolution history; (b) how Claim 4 (four-locus evolution feed) interacts with Claim 2 (FAI mechanics) at the dissolution boundary — specifically, how the per-mechanism feed structure determines which aspects of shared-substrate content flow to which evolution channels at FAI close-out; and (c) how Claim 5's recursive applicability interacts with Claim 1's perimeter-spanning construction — specifically, how meta-governance configuration (configuration-of-configuration) is handled when participating Selves have different home-perimeter governance depths.

These inter-claim synthesis notes close a gap that is structurally unavoidable in a per-sub-commitment derivation series: single-sub-commitment notes necessarily treat interactions with other sub-commitments as context rather than as the primary operational object. The inter-claim synthesis notes invert this — the interaction itself is the prior-art object.

4.2 FAI Onboarding and Offboarding

These notes formalize the operational requirements for Selves joining and leaving ongoing FAI relationships. This topic is distinct from two already-covered topics that might seem to exhaust the territory but do not: per-event withdrawal (covered earlier in Phase D2, corresponding to a Self declining participation in a specific FAI event within an ongoing relationship) and relationship exit (D2.46, covering termination of the ongoing relationship itself). Onboarding and offboarding address a third case — the governance mechanics by which a Self establishes the terms of an ongoing FAI relationship for the first time, or winds down participation in a network of relationships in a governed way that preserves home-substrate integrity and satisfies accumulated governance obligations.

The specific prior-art objects in this topic area include: initial governance-configuration alignment between Selves entering a first-time FAI relationship; the substrate-carried record required to make an ongoing relationship relationship-governed rather than event-by-event negotiated; the mechanics of onboarding a Self that has a different home-perimeter governance maturity level than existing network participants; and the offboarding sequence that ensures accumulated shared-substrate content is disposed of, transferred, or preserved according to governance-configuration terms rather than abandoned at relationship end.

4.3 Multi-Tier Evolution Mechanics

These notes formalize how FAI evolution at the inter-Self scope interacts with Paper 2's intra-Self evolution mechanisms, particularly in cases where a single FAI event generates inputs to multiple levels of a participating Self's evolution machinery simultaneously.

The prior-art gap this topic addresses is that Phase D2's Claim 4 coverage (D1.17–D1.21) establishes the four-locus evolution-feed structure and the layer-routing rule at the level of a single FAI event's hand-off. What remains is the multi-level directed selection case: when a single FAI event produces DNA-layer content for one participating Self (routing to that Self's DNA evolution channel), action-layer content for the same Self (routing to action-feedback evolution), and also constitutes a population-level pattern propagation event in the Claim 6 sense. The interaction between these simultaneous feed events — how they are ordered, whether they are governed independently or jointly, and what happens when they produce conflicting evolution signals within a single Self's home-perimeter governance — is the operational prior-art object these notes establish.

4.4 Claim 6 Remaining Operational Aspects

These notes extend Claim 6's operational prior-art coverage beyond the population-participation governance (D2.31) and pattern-propagation mechanics (D2.32) established in the second quarter.

The specific Claim 6 operational aspects targeted include: how the calibrated-humility register Paper 3 maintains for Claim 6 translates into operational governance choices (what human governance authorities must be in place before a network is operating in the Claim 6 sense rather than simply accumulating FAI events); how joint authority across population-level governance perimeters is maintained as the network grows and participating Selves' home-perimeter governance configurations diverge over time; and how the six-rung substrate-mediator ladder's sixth-rung characterization of population-scale collective evolution is operationally distinguished from multi-agent coordination systems that accumulate interaction records without the substrate-mediated, human-governed FAI machinery Claim 6 describes. This last topic is defensive-publication work against the reading that Claim 6 covers any sufficiently large collection of AI agents that interact repeatedly — a misreading the operational prior art must preempt.

5. Substantive Completeness at the Halfway Mark

The Phase D2 coverage assessment at D2.50 supports a clear finding: the operational prior-art foundation for Paper 3's six claims is substantively in place. Of the fourteen D1 sub-commitment clusters tabulated in §3, seven carry comprehensive coverage and five carry near-comprehensive coverage. Only the Claim 6 extension and cross-paper evolution-mechanics topics carry partial coverage — and those are structurally targeted by the third and fourth quarter notes rather than left as gaps.

This finding matters for understanding what D2.51 through D2.80 contribute relative to what D2.01 through D2.49 have established. The remaining 30 notes are not corrections to a foundation in progress. They add three things: depth in the near-comprehensive areas (filling the specific cross-claim interactions and multi-level evolution mechanics that the per-sub-commitment coverage necessarily left as context); population-scope operational expansion for Claim 6 (which carries the most extension-claim territory and the most important foil-differentiation work); and boundary-case precision for the Paper 2 / Paper 3 interaction points that Phase CC.2 will subsequently formalize as explicit cross-derivation prior art.

What is already in place: every core operational sequence in the Paper 3 architecture is covered. An FAI event can be traced from governance configuration through shared-substrate construction, aspect exchange under full-merge semantics, conflict handling at any tier, evolution-feed hand-off at dissolution, and governance-quality documentation — with operational prior art at each step. The governance rights that protect participating Selves' sovereignty (inspection, exit, asymmetric participation, content-tier control) are covered. The quality standards that distinguish compliant from non-compliant FAI governance are covered. The foundational commitments (instinct/reasoning separation, non-delegation, domain-agnosticism) are confirmed at the inter-Self scope. The collective-evolution network mechanics at operational level are covered for the participation-configuration and pattern-propagation dimensions.

What the remaining 30 notes add is the mortar between these established blocks: how the blocks interact when they operate simultaneously, how the architecture performs at its boundary conditions, and how Claim 6's population-scale vision cashes out in terms of specific governance obligations rather than compositional consequence. This is genuine prior-art work — inter-claim synthesis and population-scale operational specificity are patentable territory that per-sub-commitment notes leave open. But it is depth and synthesis work, not foundational coverage work. The foundation is built.

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